

UDAY JANGIR

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TECHNICAL PROFILE

Electronics & Computer Engineering student focused on embedded systems, IoT, and hardware-software integration. Experienced with ESP32-based systems, C/Python, sensor integration, Wi-Fi-enabled control, real-time data acquisition, and system-level hardware debugging through embedded and robotics projects

EXPERIENCE

intern

06/2026 – 07/2026

Railway, Signaling & Telecommunications Department

jaipur

- Gained practical exposure to railway signaling and telecommunications systems and their system-level architecture.
- Studied railway communication infrastructure and the role of networked systems in real-time railway operations.
- Observed signaling, communication, monitoring, and fault-detection systems used in railway environments.

Intern

06/2025 – 07/2025

Jaipur Metro Rail Corporation (JMRC)

Jaipur

- Diagnosed faulty SMPS units using systematic circuit tracing and hardware debugging.
- Physically mapped circuit paths and reconstructed circuit diagrams for undocumented devices.
- Studied the architecture of railway axle-counter systems used for track-vacancy detection.
- Applied structured troubleshooting and root-cause analysis to unfamiliar electronic systems.

Coordinator

2024 – 2026

Embedded Systems & Robotics Club (ESRC), MBMU

- Coordinated technical activities and collaborated with students on embedded systems and robotics projects.
- Assisted team members in troubleshooting hardware, software, and system-integration issues.
- Communicated technical concepts and supported project implementation and debugging.

EDUCATION

B.E. Electronics & Computer Engineering

2023 – Present

MBM University

Jodhpur

- CGPA: 6.0 (Current)
- Coursework: Microprocessors & Microcomputers | Real-Time & Embedded Systems | Computer Communication & Data Networks | Digital Communication Systems | Digital Electronics | Electronic Circuits

TECHNICAL SKILLS

Embedded Systems

ESP32, Microcontrollers, GPIO, PWM, Sensor Integration, Real-Time Systems

Programming

C, Python, OpenCV (Image Acquisition)

Hardware

ESP32, SMPS Repair, Multi-layer PCB Path Tracing, Circuit Reverse-Engineering.

Communication Protocols

UART, I2C, SPI

Robotics & Computer Vision

ROS 2 Humble, RViz, Gazebo, Computer Vision, ArUco

Design & Simulation Tools

Fusion 360, Proteus, Multisim, DipTrace, Renode, OpenModelica

PROJECTS

4 DOF Robotic Arm Control System

2024

C | Microcontroller | PWM | Wi-Fi | Web Interface

- Developed a control system for a 4-degree-of-freedom robotic arm supporting physical joystick and wireless web-based control. I
- Implemented microcontroller firmware for real-time input processing and PWM generation for servo positioning.
- Integrated a custom web interface over Wi-Fi for remote robotic-arm control.

Low-Cost 3D Mapping via Stereo Vision & LiDAR Fusion

Present

ESP32-CAM | Python | OpenCV | ROS 2 Humble | LiDAR

- Developing a depth estimation pipeline using dual ESP32-CAMs; calculating disparity based on baseline and focal length parameters.
- Integrating a Luna LiDAR Mini for point-cloud verification within a ROS 2 Humble environment.
- Built a custom MJPEG-to-OpenCV bridge to handle real-time image acquisition for rectification and stereo matching.

e-Yantra Robotics Competition (IIT Bombay) — Simulation

2025

Python | ROS 2 | Gazebo | PID Control | ArUco

- Movement Logic: Developed Python scripts for a Kiwi-style holonomic omnidirectional robot in Gazebo.
- Control Systems: Implemented PID controllers for precision motion, guided by simulated overhead camera feedback and ArUco marker pose estimation.
- Simulation Engineering: Studied URDF file structures, defining links, joints, and inertial elements to simulate robot kinematics accurately.

15 kg BattleBot (IIT Jodhpur)

2025

Mechanical Design | High-Torque Drive | Failure Analysis

- Fabricated a combat robot utilizing a scooter starter motor for the high-torque drive system.
- Performed post-failure analysis on the drivetrain and chassis to identify mechanical shear points during high-impact scenarios.